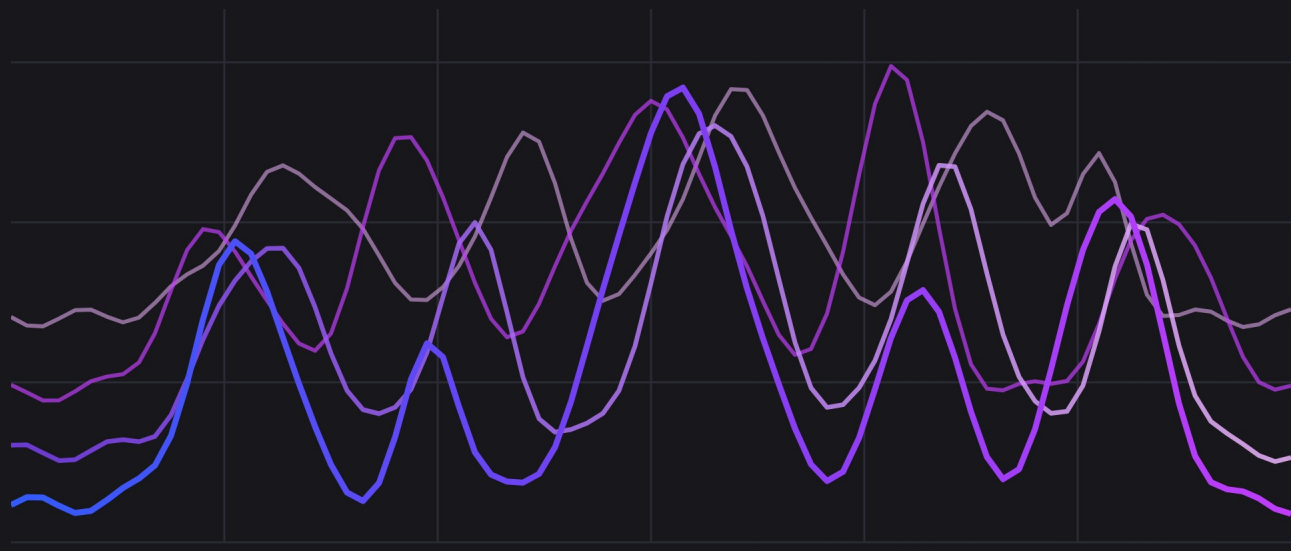
SCHUBKI NATIONAL AI HACKATHON 2026
ART FIBER SYSTEMS CHALLENGE

PRISM

An AI assistant that turns a customer's in-line spectroscopy enquiry into a grounded first technical assessment.

- Rules decide, the LLM writes
- AFS evidence, verbatim quotes
- Asks before it guesses
- Every sentence verified



github.com/m-mahmoud-mohamed/AI-Assistant-for-In-line-Spectroscopy

THE CHALLENGE

From a customer enquiry to a first technical assessment

EXAMPLE ENQUIRY FROM THE BRIEF

> “We need to monitor moisture and hydrocarbon contamination in soil. Measurements should be remote and the environment may be humid and dirty.”

Customers know **what** to monitor. They rarely know which method, spectral range, fiber probe or first test fits. An AFS expert works that out; PRISM drafts it for review.

01	**Requirements and constraints**
02	**Clarifying questions for what is missing**
03	**Recommended method(s) and spectral range**
04	**Probe and measurement concept**
05	**Reasoning, risks and uncertainty, with AFS citations**
06	**A first feasibility-study plan**

APPROACH

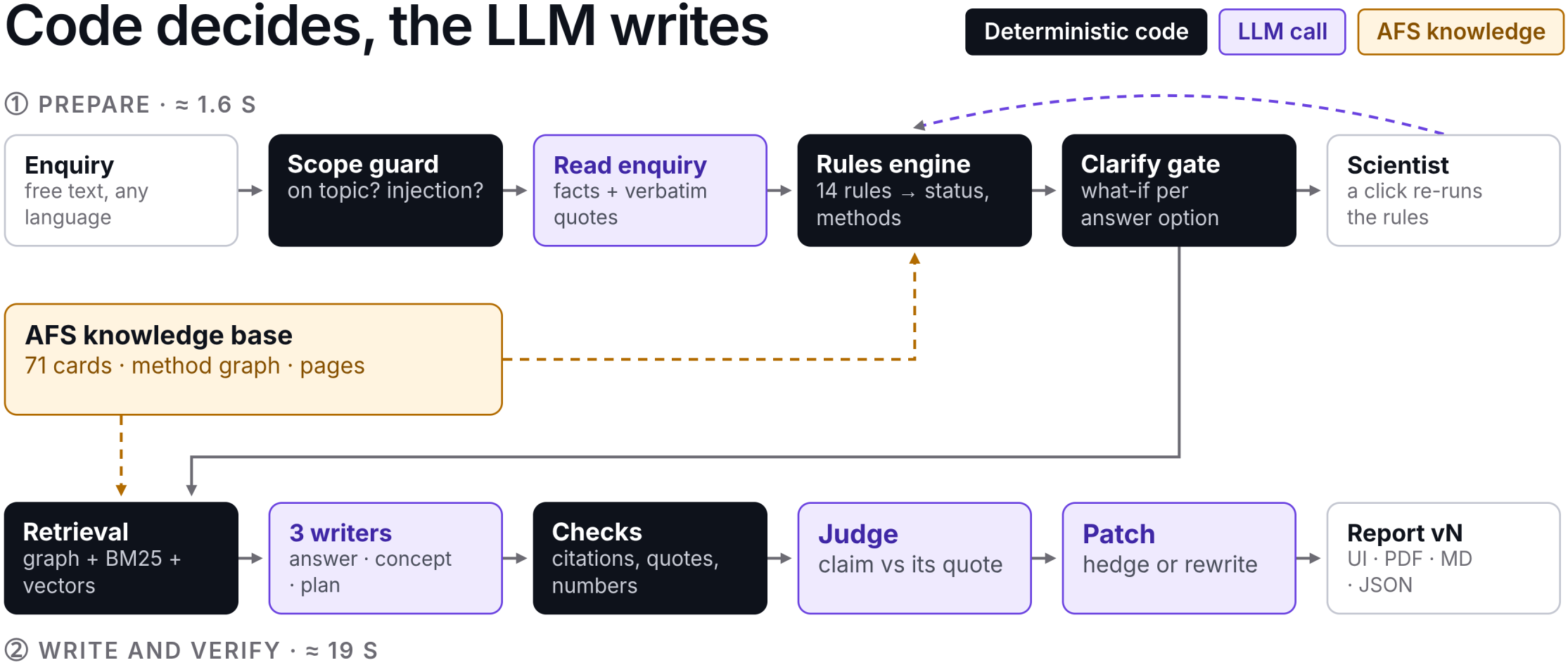
Built the other way round from a chatbot

RISK	PLAIN LLM CHATBOT	PRISM
Technical specifications	Invents plausible figures	Only from AFS evidence cards whose quote is verbatim on the cited page
Missing client facts	Fills the gaps silently	Asks first: questions ranked by how much the answer changes the recommendation
Lab or in-vitro studies	Quoted as proof of readiness	Every card carries its scope: a lab study is never proof of process performance
Confidence	Always sounds sure	Code sets the status: Provisional · Insufficient information · Outside supported scope

Rules decide **The LLM writes** **Checks verify every sentence**

SYSTEM ARCHITECTURE

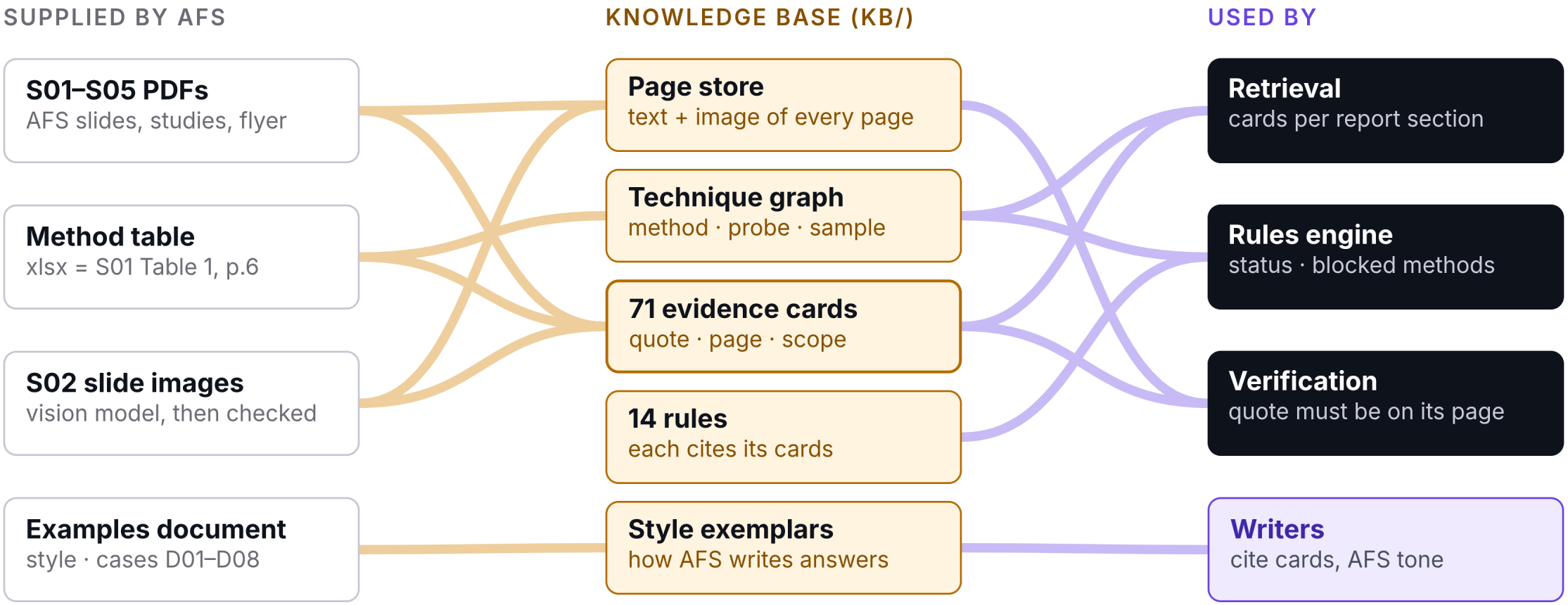
Code decides, the LLM writes



The LLM never sets the decision status: it reads, writes and judges, and can only make the status stricter.

USE OF AFS MATERIALS

Supplied materials feed every step



USE OF AFS MATERIALS

Every rule rests on cited AFS evidence cards

ANATOMY OF AN EVIDENCE CARD

S02 · page 6 | method_general

NIR · Raman · UV-Vis

QUOTE, VERBATIM

“Standard VIS-NIR Silica Fibers / Al/Cu-coated VIS-NIR Silica Fibers: 350 nm – 2400 nm”

CLAIM IT SUPPORTS

VIS-NIR silica fibers transmit about 350–2400 nm, not mid-infrared.

Checked at load: the quote is on S02 p.6

14 DETERMINISTIC RULES, BY WHAT THEY CAN DO

Gate · sets the decision status

Sample geometry · Undefined target · Clinical scope

Blocking · removes a method or fiber

Fiber window ← cites the card on the left

Premise · checks the client’s assumption

Calibration scope · Fluorophore · Evidence scope

Warning · adds a risk

Optical state · Link length · Hot or fouling · OD ≠ viability · Combination

Info · adds a note

Units (cm^{-1} vs µm) · Aqueous media

EXPLAINABILITY AND TRUST

Every sentence is checked before an expert sees it

1 · Every sentence is typed

CLIENT · AFS · EXTERNAL · ASSUMPTION · PROPOSAL · UNKNOWN

2 · Deterministic checks

card exists · quote verbatim on its page · no invented numbers

3 · Judge, one batched LLM call

is each claim supported by its own quote?

4 · Sentence patches

hedge, relabel, rewrite or delete · max. 2 rounds

5 · Report

only verified or clearly marked sentences

JUDGE VERDICTS · 122 SENTENCES



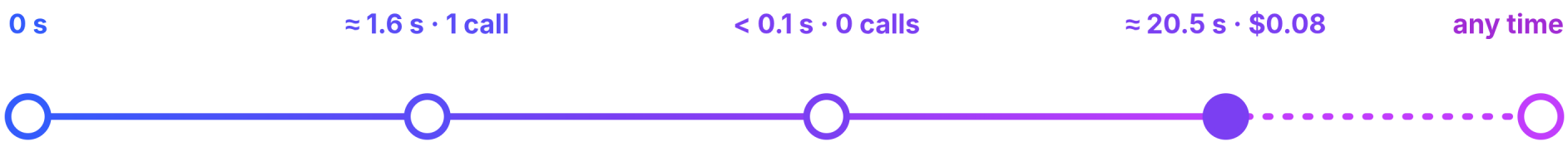
47.5 %
supported → kept

43.4 %
partly → hedged or rewritten to the quote

9.0 % unsupported → removed

THE ASSISTANT

Ask before guessing, then keep the report alive



Paste enquiry
free text, or upload your own files

Verdict preview
status and a question card, before any writer runs

Click an option
the rules engine re-runs; each option shows its effect

Report v1
verified, cited, exportable

New fact → v2
only affected sections, with “What changed”

Clarifying questions

Ranked by a what-if over the rules: only questions whose answer can change the recommendation.

Living report

A change of the primary method asks first. Every version stays readable.

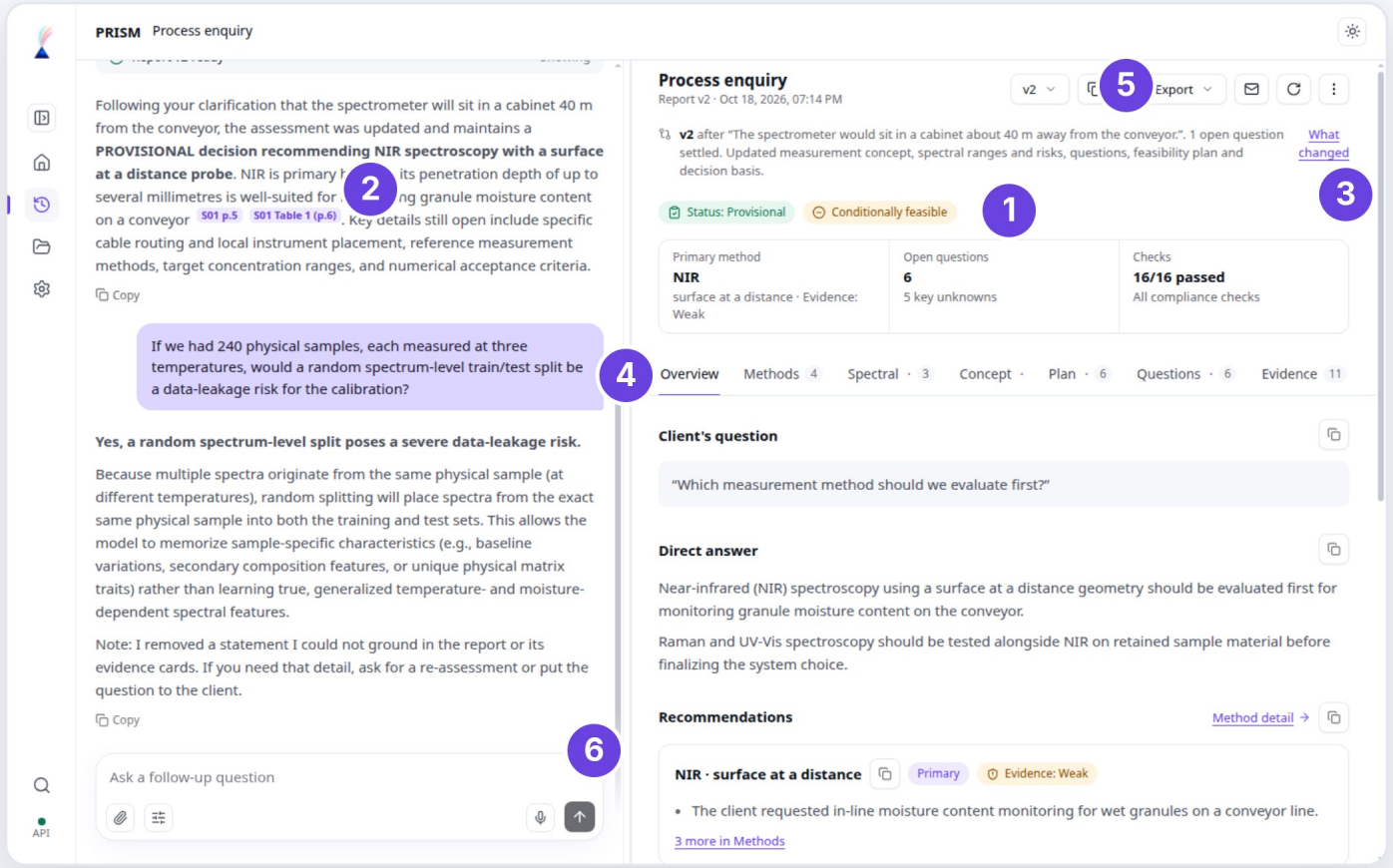
Grounded follow-ups

Calculations, e-mail drafts and tables; AFS evidence, client facts and deductions are labelled.

Times and cost measured with the OpenAI profile; the timeline is not to scale.

THE PROTOTYPE

A workspace built for reviewing, not just reading



1. Decision status and feasibility at a glance
2. Every citation opens its AFS page, highlighted
3. Report versions with "What changed"
4. Tabs: methods, spectral, concept, plan, questions, evidence, checks
5. Export A4 PDF, Markdown or JSON
6. Follow-ups answered from sources, gaps admitted

Also: own files and links cited separately from AFS · MCP tools with approvals · voice input · dark mode

Live demo: prism-spectroscopy-mahmoud-abdellahi.vercel.app

EXAMPLE RECOMMENDATION · H04

“Can the assistant approve this concept?” Not yet.

Enquiry: water content in a petroleum line by MIR ATR; analyzer in another building; reuse a silica fiber from a VIS/NIR setup.

Insufficient information | Fiber reuse: contradicted

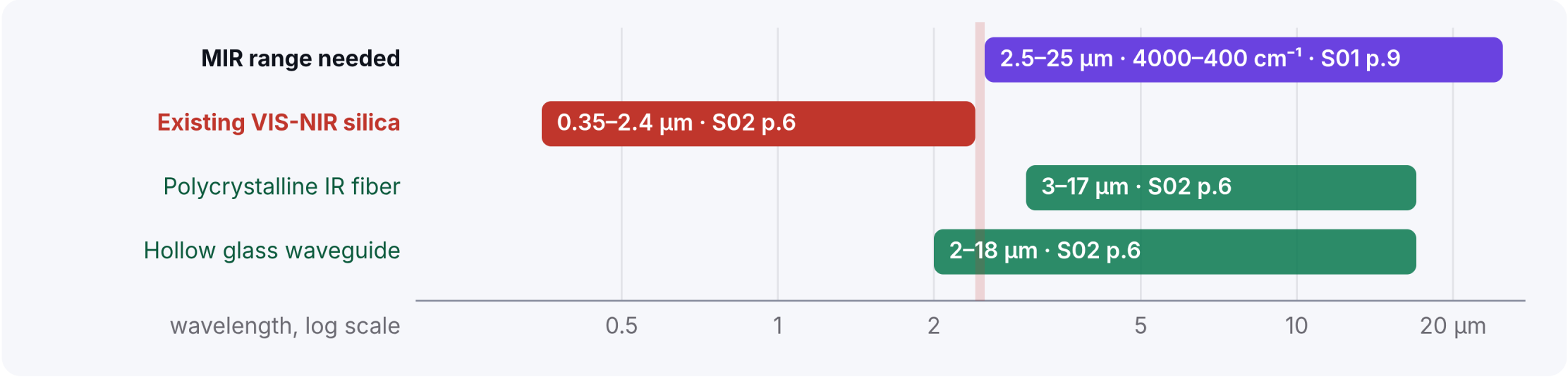
Silica stops at 2.4 µm, MIR starts at 2.5 µm: the AFS fiber data rule out the reuse. Concept: MIR ATR, conditional on a confirmed mid-IR link.

FIRST QUESTIONS TO THE CLIENT

1 · What does the liquid look like at the measurement point?

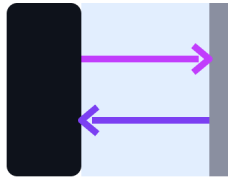
2 · Cable route and length to the analyzer?

3 · Research, pilot or production?



EXAMPLE FEASIBILITY PLAN · H02

Two dairy streams, two probe geometries, one study



CLEAR WHEY

NIR transmission or transflection

“NIR Transmission is the preferred method for transparent or semi-transparent liquids, such as skim milk, whey, or permeate.” **S01 p.18**



OPAQUE MILK CONCENTRATE

NIR diffuse reflection

“Inline NIR Reflection is primarily used for opaque or highly concentrated dairy products, such as yogurt, butter, or whole milk concentrates.” **S01 p.17**

Both: NIR 780–2500 nm **S01 p.10** · VIS-NIR silica fiber · short link next to the process · the other geometry kept as a conditional option

FIRST FEASIBILITY STUDY, SIX STEPS

1 Clarify ranges, criteria, timing, installation

2 Collect samples; reserve validation material

3 Time-matched lab protein and fat results

4 Compare interfaces and calibrated predictions

5 Validate on independent material

6 Go / no-go per stream and target

FINAL EVALUATION SCENARIOS

H01–H06: three verdicts, all generated by PRISM

H01 — Provisional

New formulation after an NIR pilot

Test the old calibration on the new blend first; “adding Raman keeps it valid” is unverified.

H02 — Provisional

Two dairy streams

Transmission for clear whey, diffuse reflection for the opaque concentrate.

H03 — Provisional

Fluorescence for glucose and biomass

Not recommended without a fluorescent target; compare NIR, Raman and UV-Vis.

H04 — Insufficient info

Silica link for MIR ATR

Fiber reuse contradicted (0.35–2.4 µm vs 2.5–25 µm); not approved.

H05 — Insufficient info

Undefined off-spec liquid

No configuration yet: a screening experiment and the deciding questions first.

H06 — Outside scope

Research sensor for surgery

An in-vitro study is not proof of clinical readiness; research-grade feasibility only.

All six pass every structural check. **make verify-outputs** replays them from the committed LLM cache and matches byte for byte, also in CI.

EVIDENCE IT WORKS

Right verdicts, no invented numbers

8 / 8
decision status as expected on D01–D08

0
invented numbers in any report

100 %
citations exist, quotes verbatim, client facts kept

12 / 12
new follow-up questions handled correctly

AGREEMENT WITH GOLD ANSWERS · D01–D08, LIVE



Traps handled: missing information, contradicted premise, fiber window, evidence scope, aqueous fluorescence.

Generality test, new domain

Polymer melt: unit conversion, hypothetical, new fact → v2, data-split advice, XRF partly outside the sources, contradicted premise, table, e-mail. Prompt injection and off-topic refused.

~590 backend and ~270 frontend tests run offline with fake models; CI on every push.

COST AND TIME PER QUERY

Fast and cheap enough to use during a customer call



< $0.01
Verdict preview
≈ 1.6 s · 1 call

$0
Answer option click
< 0.1 s · 0 calls

$0.08
Full report
≈ 20.5 s · 7 calls

≈ $0.09
Follow-up turn
4–30 s · 1–8 calls

Measured per report: 18,799 input and 4,010 output tokens (GPT-5.4 writers, 5.4-mini reader and judge, 5.4-nano patcher), priced at OpenAI list prices of mid-2026. Every call is cached, so replaying a report costs nothing.

LIMITATIONS AND NEXT STEPS

What PRISM does not do yet, and the fix for each

LIMITATION TODAY		NEXT STEP
Coverage: 71 curated cards; 129 drafted cards are unused until reviewed	→	AFS experts review the drafted cards; page-level retrieval over all AFS pages
Support strength: under half of judged claims are fully supported by their quote	→	Calibrate the judge on expert labels; the tooling is in place
Small gold set: 8 cases; 1 of 7 premise verdicts differs from the expert	→	Grow the gold set together with AFS experts
No measured spectra and no OCR of scanned pages	→	Attach measured spectra from the AFS lab to the feasibility plan
Prototype: no user accounts; enquiry text goes to the chosen LLM provider	→	Accounts and roles; an on-premise model for NDA enquiries

PRISM drafts the first assessment for an AFS expert to review; it does not replace that review.

TRY IT

PRISM drafts the first assessment. The AFS expert decides.

Live demo	prism-spectroscopy-mahmoud-abdellahi.vercel.app
Repository	github.com/m-mahmoud-mohamed/AI-Assistant-for-In-line-Spectroscopy
Answers H01–H06	outputs/ · replay with make verify-outputs